25th August Lecture-5

Concerns/suggestions from the feedback (around 30% students submitted)

a) The lectures are going slow, too slow. ≈ 2% students

b) The lectures are going fast, too fast. ≈ 1.5% students.

? [a) and b)]

c) Need more examples, practice problem: I will upload a practice set every week. But you

don't have to submit them & no solutions will be provided (the problems will be based on the examples covered in classes).

d) Students are asking irrelevant questions to earn bonus points:

- A question irrelevant for me might be meaningful for others.
- The weightage of bonus depends heavily on the quality of questions asked.

e) Due to (irrelevant) questions for bonus, the class is getting slow: Being a teacher, it is my job to ensure that every student understands the concepts thoroughly.

f) Students are not able to ask questions due to which they are losing 3-4 marks overall:

- As these marks are extra (bonus) you are

not losing anything from 100 points. You can only gain.

- It is my duty to make you all comfortable to ask questions, sometimes irrelevant (for others), but most of the time, meaningful.

g) The notes are too detailed: I want to make them self-contained.

h) Difficult to write notes:
Notes are being uploaded & you need to pay attention during lectures to access them

i) Reading notes is difficult due to small size:
Will zoom while writing.

j) Declare quizzes' dates: Not possible

Recap:

i) Propositions, conjectures, paradoxes.
ii) operations on propositions.
iii) Writing truth tables of compound propositions -

priority of logical operators

iv) Equivalent propositions

v) Conjunctive & disjunctive forms, CNF & DNF.

vi) Algebra of propositional logic.

Examples of CNFs without the truth tables.

Example 1: Convert

$u := \neg((\neg p \Rightarrow \neg q) \wedge \neg s)$ to CNF.

1. We first convert $\neg p \Rightarrow \neg q$ as follows

$$\neg p \Rightarrow \neg q \equiv \neg(\neg p) \vee \neg q$$

We know from the double negation rule that $\neg(\neg p) \equiv p$.

Hence, $\neg p \Rightarrow \neg q \equiv p \vee \neg q$.

Thus, u becomes

$\neg((p \vee \neg q) \wedge \neg r)$.

2) We should take all $\neg$ inwards.

$u \equiv (\neg(p \vee \neg q) \vee \neg\neg r)$

[$\because$ DeMorgan's law]

$\equiv (\neg p \wedge \neg\neg q) \vee \neg\neg r$

[$\because$ DeMorgan's law].

$\equiv (\neg p \wedge q) \vee r$

[$\because$ Double negation rule].

3) $(\neg p \wedge q) \vee r \equiv r \vee (\neg p \wedge q)$

[∵ commutativity of OR]

$\equiv (r \vee \neg p) \wedge (r \vee q)$

[∵ Distributivity of $\vee$ or $\wedge$].

$\equiv (\neg p \vee r) \wedge (q \vee r)$

[∵ commutativity of $\vee$].

4) In this step, we create CNFs equivalent to every OR-term in the proposition above.

$(\neg p \vee r) \equiv (\neg p \vee r) \vee F$

[∵ identity for OR]

( ) $\wedge$ ( )

$\equiv (\neg p \vee r) \vee (q \wedge \neg q)$

[∵ Contradiction for AND].

$\equiv r \vee (q \wedge \neg q)$

$\equiv (r \vee q) \wedge (r \vee \neg q)$

[∵ Distributivity of $\vee$ or $\wedge$]

$\equiv (\neg p \vee r) \vee q) \wedge$

$(\neg p \vee r) \vee \neg q)$

$\equiv [\neg p \vee (r \vee q)] \wedge$

$[\neg p \vee (r \vee \neg q)]$

[∵ Associativity of OR].

$(\neg p \vee r) \equiv (\neg p \vee r \vee q)$

$\wedge (\neg p \vee r \vee \neg q)$

Similarly, $(q \vee r) \equiv (\neg p \vee q \vee r) \wedge (p \vee q \vee r)$

$\Rightarrow (\neg p \vee r) \wedge (q \vee r)$

The final CNF is

$(\neg p \vee q \vee r) \wedge (p \vee q \vee r) \wedge (\neg p \vee r \vee q) \wedge (\neg p \vee r \vee \neg q)$

$\equiv (\neg p \vee q \vee r) \wedge (p \vee q \vee r) \wedge (\neg p \vee q \vee r) \wedge (\neg p \vee \neg q \vee r)$

$\because$ (commutativity of $\vee$).

The final CNF is

$(\neg p \vee q \vee r) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee r)$.

$\because$ Idempotence of $\wedge$

2) Convert $\neg((p \wedge q) \vee (p \wedge r)) =: u$ into a DNF.

1) $u \equiv \neg(p \wedge q) \wedge \neg(p \wedge r)$

[∵ DeMorgan's rule]

$\equiv (\neg p \vee \neg q) \wedge (\neg p \vee \neg r)$

$v \equiv \neg p \vee (\neg q \wedge r)$

2) $\equiv v \wedge (\neg p \vee \neg r)$

$\equiv (v \wedge \neg p) \vee (v \wedge \neg r)$

[∵ Distributive not an AND-term law].

$\equiv [(\neg p \vee \neg q) \wedge \neg p] \vee$

$((\neg p \vee \neg q) \wedge \neg r)$

$\hookrightarrow \equiv [(\neg p \wedge \neg p) \vee (\neg q \wedge \neg p)]$
$\vee [(\neg p \wedge \neg r) \vee (\neg q \wedge \neg r)]$

[$\because$ Distributive law]

$\equiv [\neg p \wedge \neg p] \vee (\neg q \wedge \neg p)$
$\vee (\neg p \wedge \neg r) \vee (\neg q \wedge \neg r)$

[$\because$ Associativity of $\vee$]

$\equiv \neg p \vee (\neg q \wedge \neg p) \vee$
$(\neg p \wedge \neg r) \vee (\neg q \wedge \neg r)$

$\Downarrow$ [$\because$ Idempotence of AND].

Remark: A variable with or without $\neg$ is both an

AND-term as well as an OR-term.

3) Converting the AND-terms to DNFs.

$\neg p \equiv \neg p \wedge T$ [∵ identity of AND]

$\equiv \neg p \wedge (q \vee \neg q)$ [∵ validity for OR]

$\equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q)$

[∵ Distributivity]

$w := (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (\neg q \wedge \neg p) \vee (\neg p \wedge \neg r) \vee (\neg q \wedge \neg r)$.

Homework: Convert every AND-term above into its equivalent DNF & remove

the redundant AND-terms.

Claim: Every proposition has a unique CNF & a unique DNF.

Homework: Write its proof.

27th August Lecture 6

Recap:

a) Proposition & logical operations
b) Equivalent propositions.
c) CNFs & DNFs
d) Ways to construct CNFs & DNFs:
   i) Using truth tables
   ii) Using rules of algebra of propositional logic
e) Uniqueness of CNFs & DNFs
   i) Using truth tables
   ii) Using algebra of propositions

Assignment 2

Testing equivalence of propositions using the algebra of propositional logic:

Theorem 2: Let u, v be arbitrary propositions. Then, $u \equiv v$ if and only if they have (→ the) same CNFs and DNFs.

The proof follows from the uniqueness of CNFs & DNFs.

Qne: Are conjunctive & disjunctive forms of a proposition are also unique?

Let $u = (p \vee q) \wedge (p \vee r)$

$v = (p \vee q \vee r) \wedge (p \vee q \vee \neg r)$

$\wedge (p \vee \neg q \vee r)$.

HW: Prove that $u \equiv v$

But u & v are not same.

Definition (Tautology, contradiction, and contingency):

a) A propositional formula which is true for every truth value of the propositional variables occuring in it is called a tautology.

for example, $p \vee \neg p$.

b) A propositional formula that is false for every truth value of the propositional variables appearing in it is called a contradiction.

for example, $p \wedge \neg p$.

c) A propositional formula which is neither a tautology nor a contradiction is called a contingency.

for example $p \vee q$, where p, q are propositional variables.

2. Predicate logic or First-order logic.

Propositional logic is not powerful enough to express the meanings of most of the statements in mathematics & natural languages like English for example, suppose we know that "Every student in India has studied mathematics" is true. Unfortunately, no rule of propositional logic allow us to conclude that

Shyam has studied mathematics, where Shyam is an Indian student.

Hence, we need a more powerful type of logic, called the predicate logic. Lets start with some definitions.

a) Predicates: Statements involving variables like "$x+1>5$", "student x has studied mathematics", etc. are very popular in mathematics. These statements are neither true nor false when the value of x is not specified.

In this section, we will see how to obtain propositions from such statements.

The statement $x+1>5$ has two parts $x+1$, which is called the subject & >5, which is called the predicate. The predicate refers to the property that the subject should have.

- We denote declarative statements involving a variable x by $P(x)$. Once a value is assigned to x, $P(x)$ has a

truth value & it becomes a proposition.

for example, let $P(x)$ denote $x+1>5$. Then, $P(0)$ is false where $P(0)$ means that we substitute $x=0$ in $P(x)$.

2) Quantifiers:

- Quantification: It expresses the extent to which a predicate is true over a range of values that can be substituted into the variable (or variable)

In English, the words like "all", "every", "some", etc. are used in quantification. In predicate logic, we focus on the following two types of quantification, namely universal quantification & existential quantification. Throughout this section, the domain of a predicate $P(x)$ specifies the list of values that

the variable x can take.

a) Universal quantification:
It tells us that the predicate is true for every value in the domain of the predicate.

Definition [Universal quantification]
Let P(x) be a predicate, where x is a variable. Then, the universal quantification of P(x), denoted $\forall x P(x)$, is the statement

that "$P(x)$ is true for every value of x in the domain of $P(x)$".

b) Existential quantification:

It tells us that there is at least one value in the domain for which the predicate is true.

Definition [Existential quantification]: Let $P(x)$ be a predicate & x be a variable. Then, the existential

quantification of $P(x)$, denoted $\exists x\, P(x)$, is the statement "$P(x)$ is true for at least one value of x in the domain of $P(x)$"!

- $\forall$ is called the universal operator and $\exists$ is called the existential operator.

These 2 quantifications can be denoted as follow.

Statement	When true?	When false?
$\forall x P(x)$	$P(x)$ is true for every value of x in the domain	There is a value b in the domain of $P(x)$ such that $P(b)$ is false
$\exists x P(x)$	There is a value b in the domain of $P(x)$ such that $P(b)$ is true.	$P(x)$ is false for every value in the domain of $P(x)$

Remark: Both $\forall x P(x)$ & $\exists x P(x)$ are propositions.

Examples:

i) Let $P(x)$ be the predicate "$x+1 > x$", where x is a variable. Suppose the domain of x is all the real numbers. Then,

$\forall x P(x)$ & $\exists x P(x)$

are true.

2) Let $P(x)$ be the statement $x < 2$. Suppose the domain is all the real numbers. Then $\forall x P(x)$ is false

& $\exists x P(x)$ is true.

3) Let $P(x)$ be the predicate $x > 0$. Suppose the domain is all the negative real numbers.

$\forall x P(x)$ is false

$\exists x P(x)$ is false.

Suppose that the domain is all the real numbers.

$\forall x P(x)$ is false

$\exists x P(x)$ is true.

Quantifiers over domains having finitely many elements:

Suppose the values in the domain are $a_1, \ldots, a_n$ for some natural number n. Then,

a) $\forall x\, P(x) \equiv P(a_1) \wedge P(a_2) \wedge \cdots \wedge P(a_n)$

b) $\exists x\, P(x) \equiv P(a_1) \vee P(a_2) \vee \cdots \vee P(a_n)$.

Homework: Prove these.

Remark: This equivalence does not hold if the

domain is infinite.

Precedence of quantifiers:

a) Paranthesis have the highest priority.

b) $\forall$, $\exists$ have priorities higher that all logical operators except paranthesis.

c) $\forall$, $\exists$ have the same priority & they are evaluated by first applying the innermost quantifier than moving

outwards to the left.

for example: Let $P(x,y)$ be a predicate in x, y. Then, $\forall x \exists y P(x,y)$ is interpreted as $\forall x (\exists y P(x,y))$.

1st September Lecture-7

Recap:

1) Propositional logic:
   a) Propositions & logical operations.
   b) CNFs & DNFs, and their uniqueness.
   c) Construction of CNFs & DNFs:
      i) Using truth tables
      2) Using the algebra of propositional logic.
   d) Equivalent propositions:
      i) Have the same truth tables.
      2) Have the same equivalent CNFs and the same equivalent DNFs.

c) Tautology, contradiction, contingency.

2) Predicate logic (or first-order logic):

a) Predicates, domain

b) Quantifiers: $\forall$ & $\exists$ e

c) Predicate is not a proposition but quantified predicate is a proposition.

d) Over finite domains, $\forall$ is captured by $\wedge$ and $\exists$ is captured by $\vee$.

2.1 Negating quantified expressions.

Consider the statement "Every student in the CSE department of IIITD studies discrete maths". This can be expressed logically as $\forall x\, P(x)$, where $P(x)$ means "x studies discrete maths" & the domain of x are all students in CSE in IIITD.

We want to write the negation of this statement. It is as follows:

- The rules for negation of quantifiers are called DeMorgan's Laws for quantifiers.

These are as given as

Negation	Equivalent form	when true?	when false?
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every value of x in the domain $P(x)$ is false	There exists a value of x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is a value of x for which $P(x)$ is false	$P(x)$ is true for every value of x in the domain.

How do we interpret negation of quantified statements:

a) $\neg \exists x P(x)$:

1) It is **not** the case that there **exists** a value of x in the domain for which $P(x)$ is true,

equivalently (from the table above)

2) **for every** value of x in the domain, $P(x)$ is **not** true.

b) $\neg \forall x P(x)$:

1) It is **not** the case that **for every** value of x in

the domain, $P(x)$ is true.

Equivalently (from the table above)

2) There exists a value of x in the domain for which $P(x)$ is not true.

- **Negating quantified predicates with multiple quantifiers.**

Let $P(x, y)$ be a predicate in two variables x & y. Then,

i) $\neg \forall x \forall y P(x,y) \equiv \exists x \exists y \neg P(x,y)$

2) $\neg \forall x \exists y P(x,y) \equiv \exists x \forall y \neg P(x,y)$

3) $\neg \exists x \forall y P(x,y) \equiv \forall x \exists y \neg P(x,y)$

4) $\neg \exists x \exists y \, P(x,y) \equiv \forall x \forall y \neg P(x,y)$

H.W: Prove these equivalences.

Examples:

a) Write the negation of $\forall x (x^2 > 2)$, where the domain is all real numbers.

(Predicate)

$$\neg \forall x (x^2 > 2) \equiv \exists x \neg (x^2 > 2)$$

$$\equiv \exists x (x^2 \leq 2).$$

b) Write the negation of $\exists x (x^2 = x)$, where the domain of x is all real numbers.

$$\neg \exists x (x^2 = x) \equiv \forall x \neg (x^2 = x)$$

$\equiv \forall x\,(x^2 \neq x)$.

- Translating English statements into logical expressions:

a) Convert the following into logical expression: Every student in CSE at IIITD has studied discrete mathematics.

i) Let $P(x)$ denote "x has studied discrete maths" and the domain of $P(x)$ is the students of CSE at IIITD. Then, the logical expression for the statement above

is $\forall x P(x)$.

2) We can also rewrite the statement above as "for every student x of IIITD, if x is in CSE at IIITD then x has studied DM". let $P(x)$ be "x is a student of CSE at IIITD" & $Q(x)$ be "x has studied discrete mathematics" and the domain of x is the students of IIITD. Then,

the statement given above can be logically expressed as

$$\forall x \underbrace{[P(x) \Rightarrow Q(x)]}_{\text{Predicate}}$$

- The two logical statements given above are equivalent. (Note that the domains of the two predicates given above are different).

2) Convert the following into logical expression: Some student from IIITD participated in

the International Mathematics Olympiad.

3. Proofs

A proof is a logical statement that establishes the truth of a mathematical statement.

- Recall that a proposition is a declarative statement that is true or false, but not both simultaneously.

- There are many terms used for propositions in mathematics. Some of them are:

  a) Theorem: These are the

propositions which are true.
In many cases, they have complex proofs.

b) Lemmas: These are also propositions that are true. Their proofs are usually simpler/shorter than the proofs of theorems.

c) Claims: These are also the propositions that are true. Usually, the proofs of claims are simpler/shorter than the proofs of lemmas & theorems.

d) Corollary : It is also a proposition which is true. It follows in just a few logical steps from a theorem or lemma or claim.

- Modular approach of proofs:

It means that the complex proof of a theorem/lemma is broken down into parts & each part is proven separately.

In this chapter we study various proof methods.

3.1 Logical deductions (or inference rules):

In this proof method, new propositions are proved using previously proved ones. The most important rule in this method is called **modus ponens**. This rule says the following: Suppose p, q are propositions. Then, a proof

of $\lceil p \rceil$ truth of together with a
proof of $\lceil p \Rightarrow q \rceil$ truth of is a
proof of the proposition
q.
This follows from the
truth table of $\Rightarrow$.

Example:

a) If it rains today,
then I will carry
an umbrella.

b) It is raining today.

Suppose these two statements

are true. Then, I will carry an umbrella.

As the modus ponens is very elementary and depends essentially on the proposition $p \Rightarrow q$, it may not always be possible to use modus ponens to prove propositions. Because to apply this method to prove that q is true, we need a

proposition p such that p & $p \Rightarrow q$ are true. That's why we need (more advanced) methods to prove propositions

3.2 Direct proofs.

Methodology: A direct proof of a conditional statement $p \Rightarrow q$ is constructed by first assuming that p is true & then using inference rules & the rules of predicate and propositional logic, we prove that q is also true.

- **Important properties of integers:** The following properties hold for every integers a, b, c.

1. Closure under addition an and multiplication: $a+b$, ab are integers.
2. Associativity of addition:
$$a+(b+c) = (a+b)+c.$$
3. Associativity of multiplication
$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$
4. Commutativity of addition:
$$a+b = b+a.$$

5. Commutativity of multiplication:

$$a \cdot b = b \cdot a.$$

6. Distributivity properties:

i) $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

ii) $(a + b) \cdot c = (a \cdot c) + (b \cdot c)$

This is not the same as $a + b \cdot c$.

7. Identities:

i) $0 \neq 1$

ii) $a + 0 = a$

iii) $a \cdot 0 = 0$

iv) $a \cdot 1 = a$

8. Additive inverses: There exists a unique integer m such that

$m+a=0$. In this case, $m=-1\cdot a$ & we often denote it as $-a$.

9. **Subtraction:** $a-b$ is defined as $a+(-b)$.

10. **Cancellation:** If $a\cdot b=a\cdot c$ and $a\neq 0$ then $b=c$.

Remark: There is <u>no</u> division operation for integers.

11. **No zero divisors:** If $a\cdot b=0$ then either $a=0$ or $b=0$.

12. **Signed product:**

$$-a\cdot b=a\cdot(-b)=(-a)\cdot b.$$

3rd September Lecture-8

Recap:

a) Logic:

i) Propositional logic

A) Propositions, operators

B) Equivalence of propositions

→ Truth table → CNFs & DNFs

ii) Predicate logic

A) Predicates & quantifiers

B) Negation of quantified predicates. (De Morgan's law of quantifiers).

b) Proofs:

i) Inference rule: Modus-ponens

Let p, q be propositions & suppose we have a proof of p

and a proof of $p \Rightarrow q$. This gives us a proof of q.

ii) Direct proofs: We want to prove $p \Rightarrow q$. We start by assuming that p is true & then we some inference rules and rules of propositional & predicate logic to prove q is true.

Definitions [Integers, odd & even integers]

1) The integers are $0, 1, -1, 2, -2, \ldots$

2) An integer a is said to be even if there exists an integer b such that $a = 2 \cdot b$.

for example, $0, 2, 4, 8, \ldots$
$\quad\quad\quad

3) An integer is said to odd if it is not even. Alternatively an integer a is said to be odd if there exists an integer k such that $a = 2k+1$.

Claim 3.2.1: Prove that if n is an odd integer then n^2 is also odd.

Proof: Let p be the proposition "n is an odd integer" & q be the proposition that "n^2 is an odd integers".

We are given p is true & we have to prove q is true where $p \Rightarrow q$.

Given n is an odd integer

To prove that n^2 is an odd integer

As n is an odd integer, by definition, there exists an integer k such that

$$n = 2k+1.$$

We want to show that n^2 is odd, i.e., there exists an integer m such that $n^2 = 2m+1$.

As $n = 2k+1$, we can write n^2 as

$$n^2 = n \times n = (2k+1) \times (2k+1)$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

To show that it is an integer.

$$= 2 \cdot 2k^2 + 2 \cdot 2k + 1$$

(By distributivity)

By the closure of addition &

multiplication, we get that $2k^2+2k$ is an integer. Let $m := 2k^2+2k$. Then,

$$n^2 = 2m+1.$$

By definition, n^2 is odd. Hence proved. ▨

Definition [Perfect square]: An integer a is called a perfect square if there exists an integer b such that $a = b^2$.

for example, 16 is a perfect square but 15 is not.

But $15 = (\sqrt{15})^2$.

Claim 3.2.2: Let a, b be integers. Prove that if a and b are perfect squares then $a \cdot b$ is also a perfect square.

Proof: Given a and b are perfect squares. This means that there exist integers c and d such that

$$a = c^2 \quad \text{and} \quad b = d^2.$$

We want to prove that $a \cdot b$ is a perfect square, i.e, there exists an integer e such that $a \cdot b = e^2$.

$$\begin{aligned}
a \cdot b &= c^2 \cdot d^2 \\
&= c \cdot (c \cdot d \cdot d) \\
&= c \cdot (c \cdot (d \cdot d)) \\
&= c \cdot ((c \cdot d) \cdot d) \quad [\text{Associativity of } \times] \\
&= c \cdot ((d \cdot c) \cdot d) \quad [\text{commutativity of } \times] \\
&= (c \cdot d) \cdot (c \cdot d) \quad [\text{Associativity of } \times] \\
&= (c \cdot d)^2
\end{aligned}$$

We know $c \cdot d$ is an integer by the closure property of $\times$. Let $e := c \cdot d$. Hence, $a \cdot b = e^2$. ∎

Claim 3.2.3: Let a, b, c be integers.

If $a \neq 0$ and $a \cdot b = a \cdot c$ then prove that $b = c$.

Remark: As division is invalid for integers, the above claim requires a proof. We will prove it using properties of integers other than the cancellation rule.

Proof: $\overrightarrow{\text{Given}}$ $a \cdot b = a \cdot c$ and $a \neq 0$

$$a \cdot b = a \cdot c$$

$$\Rightarrow (a \cdot b) + (-(ac)) = (a \cdot c) + (-(ac))$$

[$\because$ Additive inverse]

$$(a \cdot b) + (a \cdot (-c)) = 0$$

[$\because$ additive inverse & signed product]

$$ab + a.(-c) = 0$$

$$a[b + (-c)] = 0$$

[∵ Distributivity].

$$a[b - c] = 0$$

[∵ subtraction]

$\Rightarrow$ either $a = 0$ or $(b - c) = 0$

[∵ No zero divisors]

Given $a \neq 0$. Hence $b - c = 0$.

$$b + (-c) = 0$$

$$b + (-c) + c = c$$ [∵ additive inverse of c]

$$b = c$$ [∵ additive inverse & subtraction] ∎

Notation: Let a, b be integers. Then, we say $a \mid b$ if and only if there is an integer c such that

$$b = a \cdot c.$$

For example, $4 \mid 8$ because

$$8 = 4 \cdot 2.$$

Claim 3.2.4: Let a, b, c be integers. If $a \mid b$ and $b \mid c$ then prove that $a \mid c$.

Homework: Prove it.

Remark: Don't confuse $a \mid b$ with the fraction $\frac{a}{b}$.

Homework: Prove that for every integer a there is a <u>unique</u> integer b such that $a + b = 0$.

3.3 Proof by contraposition:

We know that $p \Rightarrow q$ & $\neg q \Rightarrow \neg p$ (also called the contrapositive) are equivalent.

Claim 3.3.1: Let a, b, n be positive integers. Prove that if $n = a \cdot b$ then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Homework: Try to prove this using direct proofs.

Proof [By contraposition]: Let p be the proposition "$n = ab$" and q be the proposition "$a \leq \sqrt{n}$ or $b \leq \sqrt{n}$". We want to prove $p \Rightarrow q$, which is equivalent to $\neg q \Rightarrow \neg p$.

$\neg p$: $n \neq ab$, i.e. $n < ab$ or $n > ab$.

$\neg q$: $a > \sqrt{n}$ and $b > \sqrt{n}$.

We are given that $\neg q$ is true, i.e. $a > \sqrt{n}$ and $b > \sqrt{n}$. This means that

$$a = \sqrt{n} + \varepsilon$$
$$b = \sqrt{n} + \delta,$$

where ε, δ are real numbers $\varepsilon > 0, \delta > 0$.

$$a \cdot b = (\sqrt{n} + \varepsilon) \cdot (\sqrt{n} + \delta)$$
$$= n + \sqrt{n}(\varepsilon + \delta) + \varepsilon\delta.$$

Because $\varepsilon > 0, \delta > 0$, we have

$$\varepsilon + \delta > 0 \quad \text{and} \quad \varepsilon\delta > 0.$$

$$\Rightarrow \sqrt{n}(\varepsilon + \delta) + \varepsilon\delta > 0$$

$$\Rightarrow a \cdot b > n.$$

$$\Rightarrow n < ab.$$

$$\Rightarrow n \neq ab.$$

Hence $\neg p$ is true.

We have shown $\neg q \Rightarrow \neg p$,

which is equivalent to $p \neq q$.

Hence proved. ∎

Lecture-9

Recap:

a) Rule of inference : modus ponens

b) Direct proofs : various properties of integers

c) Proof by contraposition: power of contrapositive

Some important announcements

1) Mid-sem exam : 27th September (Saturday)

Duration: 2.5 hours - from 1:30 PM - 4:00 PM

2) Syllabus: Till proof by

contradiction.

3) feedback (quiz): 60 students

a) (n-1) policy: Not possible!

b) long: you will get used to it.

c) Not to include the content covered on the day of quiz: I am preparing you all for any type of question in DM for the rest of your life.

d) Hard, very hard: It was not. Wait for other quizzes.

- Feedback on lectures:
  a) Surprise attendence: Not in the course's evaluation
  b) Pace slow (~5% of feedback): Have patience
  c) Writing in latex is a problem: It is optional
  d) Irrelevant queries being asked: Most welcome

3.4 Proving Bi-conditionals (or if and only if statements)

Let p, q be propositions. Suppose we want to prove that p is

true if and only if q is true (in short, $p \iff q$ or p iff q). We have to prove two things here:

a) $p \Rightarrow q$ b) $q \Rightarrow p$.

Claim 3.4.1: An integer n is even if and only if n^2 is even.

Proof: Let p & q be the propositions "n is even" and "n^2 is even", respectively

We have to prove p is true if and only if q is true. This is the same as proving the following:

a) $p \Rightarrow q$ b) $q \Rightarrow p$.

a) $p \Rightarrow q$: Given that n is even. By definition, there exists an integer k such that $n = 2 \cdot k$

We want to prove that n^2 is even, which means there exists an integer

m such that $n^2 = 2 \cdot m$.

Homework: Complete the proof of part a).

b) $q \Rightarrow p$. Given that n^2 is even. We want to show that n is even.

$\neg p \Rightarrow \neg q$: If n is odd then n^2 is odd.

$\equiv$

$q \Rightarrow p$.

$\blacksquare$

Remark: A hybrid proof is a proof method that uses more than 1 proof methods. The proof

of Claim 3.4.1 is an example of hybrid proof.

3.5 Proof by counterexample:

This method is often used when want to prove that some proposition is false.

Claim 3.5.1: Every integer greater than 10 is an even number?

Proof: This is false as $11 > 10$ & 11 is odd. ∵ $11 = 2 \cdot 5 + 1$ ■

3.6 Proof by cases:

In this technique, a complicated proof is broken down into various cases & each case is proven separately. The important thing in this method is that all the possible cases must be proven.

Claim 3.5.1 : For every integer a, $a(a-1)$ is even.

Proof: $\forall a$ ($a \cdot (a-1)$ is even) where the domain of a is

integers. Let us pick an integer a arbitrarily. Then, we have to show that

$$a \cdot (a-1) = 2 \cdot k$$

for some integer k.

There are two possibilities:

a) a is even

b) a is odd.

Case 1: a is even: We have to show that $a(a-1)$ is even. Complete it (HW)

Case 2: a is odd: We want to show that $a(a-1)$ is even.

Complete the proof for this case (HW). ∎

Terminology: Given any two people, there are two possibilities: either they have met or they have not met. If every pair in a group of people has met then this group is called a **club**. If every pair in this group has not met then is called a **group of strangers**

Claim 3.5.2: Prove that every group of six people includes either a clique of three people or a group of three strangers.

Proof: Suppose there are 6 people say

$A_1, A_2, A_3, A_4, A_5, A_6$.

There are two cases

a) A_1 has met atleast three of $A_2, \cdots, A_6$

b) A_1 has met at most two of $A_2, \cdots, A_6$.

15th Sept, 25 Lecture-10

Recap:

1) Proof by inference rule - Modus Ponens
2) Direct proofs
3) Proof by contraposition
4) Proving bi-conditionals
5) Proof by counter example
6) Proof by cases

Claim 3.5.2: Prove that every group of six people includes either a club of three people or a group

of three strangers.

Proof: Suppose there are six people $A_1, \ldots, A_6$. There are two cases here:

a) A_1 has met at least three people among $A_2, \ldots, A_6$

b) A_1 has not met at least three people among $A_2, \ldots, A_6$.

Case 1: Suppose A_1 met at least three people among $A_2, \ldots, A_6$. Then there are two sub-cases here.

Subcase 1.1: At least two of these three people have met. Then, these two people

along with A_1 form a club of three people.

Subcase 1.2: No pair of people in the group of these three people have met. Then, these three people form a group of 3 strangers.

This proves Case 1.

Case 2: Suppose A_1 has not met three people among $A_2, \ldots, A_6$. Then there are 2 possible cases.

Subcase 2.1: (Every pair of people among these three people have met) This forms a club of 3 people.

Subcase 2.2: There exists a pair of people among these three people who have not met. Then, these 2 people along with A_1 forms a group of 3 strangers.

This proves Case 2 & it along with Case 1 proves the claim. ∎

7. Proof by contradiction.

In this methodology, we assume that the given proposition is false. Then, we eventually show that either some false proposition becomes true or some true proposition becomes false. Since this can't happen (as no proposition is simultaneously true and false), our assumption is incorrect and the given proposition must be true.

Definition [Rational numbers]: A number a is said to be rational if there exist integers b, c such

that $c \neq 0$ and $a = \frac{b}{c}$. $\exists b \; \exists c$ $(a = \frac{b}{c})$

for example: $\frac{17}{42}$, $\frac{15}{3}$ are rational number.

domain of b is integers domain of c are non zero integers

Observation: Every integer is a rational number.

Homework.

Definition [Irrational number]: A number a is said to be irrational if it is not a rational number. Equivalently, for every integers b, c satisfying $c \neq 0$, $a \neq \frac{b}{c}$.

Claim 3.7.1: Prove that there do not exist integers a, b such that $18a + 6b = 1$.

Proof: The given proposition is

$$\neg \exists a \exists b \, (18a + 6b = 1),$$ where the domains of a, b are integers.

This proposition is equivalent to

$$\forall a \forall b \, (18a + 6b \neq 1).$$

Let us call this proposition p.

We assume that p is false, i.e., $\neg p$ is true, which means there exist integers a, b such that

$$18a + 6b = 1.$$

By the distributivity property,

$$18a + 6b = 6(3a + b).$$

Hence, $6(3a + b) = 1.$

We treat a, b, 3, 6, & 1 as rational numbers $\frac{a}{1}, \frac{b}{1}, \frac{3}{1}, \frac{6}{1}$ & $\frac{1}{1}$.

Then, the above equation looks

$$\frac{6}{1}\left(\frac{3}{1}\cdot\frac{a}{1}+\frac{b}{1}\right)=\frac{1}{1}$$

We divide both sides by 6 and get

$$\left(\frac{3}{1}\cdot\frac{a}{1}+\frac{b}{1}\right)=\frac{1}{6}$$

As $\frac{3}{1}, \frac{a}{1}, \frac{b}{1}$ are integers, we write them as 3, a, b.

Then,

$$3a+b=\frac{1}{6}.$$

As 3, a, b are integers, we

know from the closure properties of integers that $3a+b$ is an integer. But $\frac{1}{6}$ is not an integer as $0 < \frac{1}{6} < 1$, and there is no integer between 0 & 1. Hence, $3a+b$ can never be equal to $\frac{1}{6}$. Hence, our assumption was incorrect. Thus, the proposition p is true. □

Remark: We know that division is not allowed in integers, but allowed for rational numbers.

Thus, whenever we want to perform division on integers, we treat them as rational numbers.

Claim 3.7.2: Prove that the sum of a rational and an irrational number is always irrational.

Proof: Let p be the proposition

$$\forall a \, \forall b \, (a + b \text{ is irrational}),$$

where the domain of a is all

rational numbers & the domain of b is all irrational numbers. we assume for contradiction that p is false, i.e., $\neg p$ is true. By DeMorgan's laws,

$$\neg p \equiv \exists a \exists b \text{ (} a+b \text{ is rational)}$$

Then, there exist integers c, d, $d \neq 0$ s.t.

$$a + b = \frac{c}{d}.$$

As a is rational, there exist integers e, f s.t. $f \neq 0$ and

$$a = \frac{e}{f}. \text{ Then,}$$

$$\frac{e}{f} + b = \frac{c}{d}.$$

additive inverse of $\frac{e}{f}$

On subtracting $\left(-\frac{e}{f}\right)$ on both sides, we get

$$b = \frac{c}{d} - \frac{e}{f}.$$

On making the denominator same on the R.H.S, we get

$$b = \frac{cf - ed}{df}$$

] Integer

] Integer $\neq 0$

By the closure properties of integers, we know $cf - ed$ & df are integers. Moreover, as both d & f are non-zero,

it follows from the no-zero divisors property that $df \neq 0$. Thus, b is a rational number. This is a contradiction. Hence, the proposition p is true. ∎

Definition [Co-prime integers]: We say that two integers a, b are co-prime if there doesn't exist an integer $c > 1$ s.t.

$c \mid a$ & $c \mid b$.

$a = c \cdot a'$, a' integer. $b = c \cdot b'$, b' integers.

for example, 9 & 10 are co-prime.

Claim 5.7.3: Prove that $\sqrt{2}$ is irrational.

Proof: Assume for the contradiction that this is not true, i.e., $\sqrt{2}$ is rational. Then, there exist integers $a, b, b \neq 0$ such that

$$\sqrt{2} = \frac{a}{b}.$$

We can assume without loss of generality that a & b are co-prime. This is so because if there exists an integer $d > 1$ s.t. $d \mid a$ & $d \mid b$ then d gets cancelled

Homework: using properties of integers

in $\frac{a}{b}$. Then, $\sqrt{2} = \frac{a}{b}$ can be written as

$$\sqrt{2}b = a$$

Now, take square of both sides. Then,

$$2b^2 = a^2.$$

Thus $2 | a^2$ & we know that this means $2 | a$. Then, $a = 2k$ for some integer k & $a^2 = 4k^2$. On substituting this in

the equation above, we get

$$2b^2 = 4k^2$$

which is the same as

$$2(b^2 - 2k^2) = 0$$

[by using additive inverse & distributing]

By the no non-zero divisors property, we get

$$b^2 - 2k^2 = 0$$

which is the same as

$$b^2 = 2k^2.$$

This implies $2 \mid b^2$, which

implies $2|b$. Hence, we have shown that $2|a$ & $2|b$. But a & b are co-prime & this can never happen. This is a contradiction. Hence, $\sqrt{2}$ is irrational.

∎

17th Sept Lecture-11

Recap

1) Proof by inference rule
2) Direct proofs
3) Proof by contraposition
4) Proving bi-conditionals
5) Proof by counterexamples
6) Proof by cases
7) Proof by contradiction.

Definition [Prime numbers]: A positive integer a is called a prime number if $a > 1$ and

the only positive integers that divide a are 1 and a.

Definition [finite & infinite]: We say that a collection of elements is finite if the total number of elements in this collection is n for some natural number n. If such an n does not exist, then this collection is said to be infinite.

For example the collection of integers between 0 to 100 is finite but the collection

of all integers is infinite.

Claim 3.7.4: Prove that there are infinitely many prime numbers.

Proof: Assume for contradiction that there are finitely many prime numbers, say $p_1, \dots, p_n$ for some natural number n. Let p_{n+1} be defined as

$$p_{n+1} := p_1 p_2 \cdots p_n + 1.$$

Assume $p_1 < p_2 < \cdots < p_n \Rightarrow p_{n+1} > p_n$.

By the closure properties of integers, it follows that p_{n+1} is a positive integer. We will show that p_{n+1} is also a prime number, which would contradict our assumption that there are only n prime numbers.

Suppose p_{n+1} is not a prime number. The following fact, which we will prove later will be helpful.

$p_{n+1} = p_i \cdot k$

$p_1 \cdots p_n + 1 = p_i \cdot k.$

Fact 1: Let a be an integer which is not a prime number. Then, there exists a prime number b such that $b < a$ and $b \mid a$.

We assume this fact to complete our proof. As per our assumption, p_{n+1} is not a prime number. Then there exists a prime number b such that $b < p_{n+1}$ & $b \mid p_{n+1}$ (By fact 1). As $p_1, \ldots, p_n$ are the only

prime numbers as per our assumption, we get that there exists a prime number p_i such that $p_i \mid p_{n+1}$, where $1 \leq i \leq n$. This means

$$p_{n+1} = p_i \cdot k$$

for some integer k. This implies

$$p_1 \cdots p_n + 1 = p_i k$$

$$\Rightarrow p_i \left(k - p_1 \cdots p_{i-1} p_{i+1} \cdots p_n\right) = 1$$

[By using additive inverse & distributivity]

Treating everything here as rational numbers & dividing by p_i, we get

$$k - p_1 \cdots p_{i-1} p_{i+1} \cdots p_n = \frac{1}{p_i}$$

we know from the closure properties of integers that

$$k - p_1 \cdots p_{i-1} p_{i+1} \cdots p_n \text{ is}$$

an integer. But $\frac{1}{p_i}$ is not an integer as $0 < \frac{1}{p_i} < 1$ & there is no integer between 0 & 1. Here we are using the

fact that p_i is a prime number and hence $p_i > 1$.

Thus we get a contradiction.

Hence, there are infinitely many prime numbers. ∎

Homework: Prove that for every integer n, 4 does not divide (n^2+2).

4. Set Theory

Set theory is the basement of mathematics. After the introduction of set theory in the 19th century by Georg Cantor, mathematics has seen tremendous developments.

Definition [Set]: A set is a collection of elements. The elements can be anything like numbers, people, computers, etc.

- Sets are usually denoted by

upper-case Roman letters like A, B, etc.

- A conventional way to write a set is to list all the elements inside curly brackets ({}).

This representation is fine if the number of elements in a set is finite. Some examples in this representation are

1) $A = \{1, 2, 3\}$.

2) $B = \{$Ram, Madhav, Chaitanya$\}$

3) $C = \{\{1,2\}, \{2,3\}\}$ – this is an example of a sets whose elements

are also sets.

- Some important properties of sets:

1) In every set, elements are not written in any order. This means $\{a, b\}$ & $\{b, a\}$ are the same sets.

2) In every set, each element can appear exactly once. If there is a repetition, we just keep one of them. Thus, $\{x, x\}$ is the set $\{x\}$.

3) The notion $a \in A$ means that a is an element of A & $a \notin A$ means that a is not an element of A. For example, $5 \in \mathbb{Z}$ $\frac{1}{2} \notin \mathbb{Z}$.

- We now present some important and popular sets in mathematics:

1) ϕ: It is called the empty set, i.e. the set with no elements. Hence,

$$\phi = \{\}.$$

2) $\mathbb{N}$: The set of natural numbers, i.e.,

$\mathbb{N} = \{0, 1, 2, \ldots\}$

3) $\mathbb{Z}$: The set of integers,
i.e., $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$

4) $\mathbb{Q}$: The set of rational numbers.

5) $\mathbb{R}$: The set of real numbers

6) $\mathbb{C}$: The set of complex numbers.

- Another way to denote a set uses predicates & has

a much concise representation.
In particular, let $P(x)$ be a predicate in x the variable.
Then $\{a : a$ is in the domain of x and $P(a)$ is true$\}$.

for example, read as such that

$$A = \{x \in \mathbb{N} \mid \exists m \in \mathbb{N} \text{ s.t. } (x = 2 \cdot m)\}.$$

Then A is a set of even numbers. Unlike the previous representation, this one handles infinite sets.

Passing criteria:

Score $\geq$ 30% of [Quizes (20%) + Mid-sem (30%) + End-sem (38%)]

$\hookrightarrow$ 88 points.

26.4 marks.

26.64 F

Few points

i) 2:30 hrs : 1:30 pm – 4:00 pm.

2) You must use either a black or a blue pens.

3) You will get extra sheets.

4) Syllabus: till proof by contradictio